

between various development and deployment environments is ensured with the aid of Docker.

Kubernetes: An open source platform for container orchestration that streamlines the deployment, scalability, and administration of applications within containers.

Terraform: This open source infrastructure as code (IaC) solution lets you use a high-level configuration language to define and provision infrastructure.

Ansible: An open source automation platform for task automation, application deployment, and configuration management. It employs an easy-to-read YAML syntax.

Chef: A configuration management tool that uses a Ruby-based domain-specific language to automate the management and configuration of servers.

GitHub Actions: This tool allows continuous integration and delivery (CI/CD) workflows within the platform, which facilitates the automation of testing, building, and deployment procedures.

Azure DevOps: A collection of Microsoft development tools for managing the full DevOps lifecycle that includes Azure Pipelines, Azure Repos, Azure Test Plans, and Azure Artifacts.

Career trends in DevOps for 2024

Increased demand for DevOps skills:

The demand for professionals with DevOps skills continues to grow as more organisations adopt DevOps

practices to improve their software development and delivery processes.

Cloud-native technologies:

Proficiency in cloud platforms like AWS, Azure, and Google Cloud, as well as cloud-native tools like Kubernetes and Docker, is increasingly important. The shift towards cloud-native architectures is driving demand for expertise in these technologies.

Security integration (DevSecOps):

Integrating security practices into DevOps processes, known as DevSecOps, is becoming a priority. Professionals with skills in security automation, vulnerability management, and compliance are in high demand.

Certification and continuous learning:

Obtaining relevant certifications (e.g., AWS Certified DevOps Engineer, Kubernetes Certified Administrator) and engaging in continuous learning are becoming standard practices to stay competitive in the field.

DevOps in emerging industries:

DevOps practices are expanding beyond traditional tech companies into industries such as healthcare, finance, and retail. Professionals with domain-specific knowledge combined with DevOps skills will find new opportunities.

Key roles and responsibilities of a DevOps professional

Continuous integration and continuous deployment (CI/CD)

- Developing and maintaining CI/CD pipelines to automate the build, test,

and deployment processes.

- Ensuring code changes are tested and integrated into the main branch frequently and reliably.

Infrastructure management

- Provisioning and managing infrastructure using infrastructure as code (IaC) tools such as Terraform or Ansible.
- Monitoring and optimising infrastructure performance, scalability, and reliability.

Monitoring and performance management

- Implementing monitoring and logging solutions to track application and infrastructure performance.
- Analysing metrics and logs to identify and resolve issues proactively.

Troubleshooting and incident management

- Diagnosing and resolving issues related to application deployments, infrastructure, and performance.

To sum up, DevOps is a methodology every aspiring software engineer must get acquainted with, as it is in demand even as many other skills are getting redundant. **END** 🐧

References

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DevOps professional payroll and role types

- Entry-level DevOps engineer: Salary range: \$60,000 - \$85,000 per year
- Mid-level DevOps engineer: Salary range: \$85,000 - \$120,000 per year
- Senior DevOps engineer: Salary range: \$120,000 - \$160,000 per year
- DevOps architect: Salary range: \$140,000 - \$180,000 per year
- DevOps manager/director: Salary range: \$150,000 - \$200,000+ per year
- Cloud DevOps engineer: Salary range: \$100,000 - \$150,000 per year

Note: Salaries can vary based on factors such as location, company size, and industry. For instance, tech hubs like San Francisco, New York, and Seattle may offer higher salaries compared to other regions.